

Cauchy Limit Examples.

① Prove $\lim_{x \rightarrow 8} \frac{3x}{x-2} = 4$.

Let $\varepsilon > 0$ be any positive real number.

Begin by assuming $|x-8| \leq 1$ so $7 \leq x \leq 9$. In this case

$$|f(x) - L| = \left| \frac{3x}{x-2} - 4 \right| = \left| \frac{3x - 4x + 8}{x-2} \right| = \left| \frac{-x+8}{x-2} \right| = \left| \frac{-1}{x-2} (x-8) \right| = \left| \frac{-1}{x-2} \right| |x-8|$$

Now the max value of $\left| \frac{-1}{x-2} \right|$ when $7 \leq x \leq 9$ is $\frac{1}{5}$

So if $|x-8| \leq 1$ we know $|f(x) - L| = \left| \frac{-1}{x-2} \right| |x-8| \leq \frac{1}{5} |x-8| < \varepsilon \Rightarrow |x-8| < 5\varepsilon$

Choose $\delta = \min \{1, \text{or } 5\varepsilon\}$.

$\therefore \forall \varepsilon > 0$ if $0 < |x-8| < \delta$ we have $|x-8| < 1$ and $|x-8| < 5\varepsilon$

$$\text{so } \left| \frac{3x}{x-2} - 4 \right| = \left| \frac{-1}{x-2} \right| |x-8| < \frac{1}{5} \cdot 5\varepsilon = \varepsilon$$

$$\therefore \lim_{x \rightarrow 8} \frac{3x}{x-2} = 4$$

② Prove $\lim_{x \rightarrow 3} \sqrt{5x+10} = 5$.

Let $\varepsilon > 0$ be any positive real number.

Assume $|x-3| \leq 1$ so $2 \leq x \leq 4$.

$$|f(x) - L| = \left| \sqrt{5x+10} - 5 \right| = \left| \frac{(\sqrt{5x+10} - 5)(\sqrt{5x+10} + 5)}{\sqrt{5x+10} + 5} \right| = \left| \frac{5x-15}{\sqrt{5x+10} + 5} \right| = \left| \frac{5}{\sqrt{5x+10} + 5} \right| |x-3|$$

The max value of $\left| \frac{5}{\sqrt{5x+10} + 5} \right|$ when $2 \leq x \leq 4$ is $\frac{5}{\sqrt{20} + 5}$

So if $|x-3| \leq 1$ we know $|f(x) - L| = \left| \frac{5}{\sqrt{5x+10} + 5} \right| |x-3| \leq \frac{5}{\sqrt{20} + 5} |x-3| < \varepsilon \Rightarrow |x-3| < \frac{\sqrt{20} + 5}{5} \varepsilon$

Choose $\delta = \min \left\{ 1, \text{or } \frac{\sqrt{20} + 5}{5} \varepsilon \right\}$

$\therefore \forall \varepsilon > 0$ if $0 < |x-3| < \delta$ we have $|x-3| < 1$ and $|x-3| < \frac{\sqrt{20} + 5}{5} \varepsilon$

$$\text{so } \left| \sqrt{5x+10} - 5 \right| = \left| \frac{5}{\sqrt{5x+10} + 5} \right| |x-3| < \frac{5}{\sqrt{20} + 5} \cdot \frac{\sqrt{20} + 5}{5} \varepsilon = \varepsilon$$

$$\therefore \lim_{x \rightarrow 3} \sqrt{5x+10} = 5.$$